Fourier cosine series

We reviewed even and odd functions when we first introduced Fourier series, but remember again that an **even function** $f(x)$ is symmetric about the y-axis and satisfies $f(-x) = f(x)$, while an **odd function** $f(x)$ is symmetric about the origin and satisfies $f(-x) = -f(x)$.

The cosine series

If we try to find the Fourier series representation,

$$f(x) = A_0 + \sum_{n=1}^{\infty} A_n \cos\left(\frac{n\pi x}{L}\right) + \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right)$$

$$A_0 = \frac{1}{2L}\int_{-L}^{L} f(x)\ dx$$

$$A_n = \frac{1}{L}\int_{-L}^{L} f(x)\cos\left(\frac{n\pi x}{L}\right)\ dx \qquad n = 1,\ 2,\ 3,\ \ldots$$

$$B_n = \frac{1}{L}\int_{-L}^{L} f(x)\sin\left(\frac{n\pi x}{L}\right)\ dx \qquad n = 1,\ 2,\ 3,\ \ldots$$

of an even function on $-L \leq x \leq L$, we'll discover that the coefficients B_n are 0, such that the Fourier series of the even function will simplify to just

$$f(x) = A_0 + \sum_{n=1}^{\infty} A_n \cos\left(\frac{n\pi x}{L}\right)$$

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$$A_0 = \frac{1}{L}\int_0^L f(x)\ dx$$

$$A_n = \frac{2}{L}\int_0^L f(x)\cos\left(\frac{n\pi x}{L}\right)\ dx \qquad n = 1, 2, 3, \ldots$$

Because this is a Fourier series in terms of only cosine functions, we call it a **Fourier cosine series**. What this means is that, when we want to find the Fourier series representation of an even function, we actually only need to find its Fourier cosine series. If we know the function is even, we don't have to bother computing B_n or trying to find the sine part of the Fourier series representation.

Let's do an example with an even function to prove that this is true.

Example

Find the Fourier series and Fourier cosine series of $f(x) = x^2$ on $-L \leq x \leq L$, and show that they are equivalent.

Let's use the original definition of a Fourier series, which means we'll start by calculating A_0, A_n, and B_n. For A_0, we get

$$A_0 = \frac{1}{2L}\int_{-L}^L f(x)\ dx$$

$$A_0 = \frac{1}{2L}\int_{-L}^L x^2\ dx$$

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$$A_0 = \frac{1}{6L}x^3\Big|_{-L}^{L}$$

$$A_0 = \frac{1}{6L}L^3 - \frac{1}{6L}(-L)^3$$

$$A_0 = \frac{1}{6}L^2 + \frac{1}{6}L^2$$

$$A_0 = \frac{L^2}{3}$$

For A_n, we get

$$A_n = \frac{1}{L}\int_{-L}^{L} f(x)\cos\left(\frac{n\pi x}{L}\right)\,dx$$

$$A_n = \frac{1}{L}\int_{-L}^{L} x^2\cos\left(\frac{n\pi x}{L}\right)\,dx$$

Use integration by parts with $u = x^2$, $du = 2x\ dx$, $dv = \cos(n\pi x/L)\ dx$, and $v = (L/n\pi)\sin(n\pi x/L)$.

$$A_n = \left[\frac{x^2}{n\pi}\sin\left(\frac{n\pi x}{L}\right) - \frac{2}{n\pi}\int x\sin\left(\frac{n\pi x}{L}\right)\,dx\right]\Big|_{-L}^{L}$$

Use integration by parts with $u = x$, $du = dx$, $dv = \sin(n\pi x/L)\ dx$, and $v = -(L/n\pi)\cos(n\pi x/L)$.

$$A_n = \left[\frac{x^2}{n\pi}\sin\left(\frac{n\pi x}{L}\right) + \frac{2L}{(n\pi)^2}\left[x\cos\left(\frac{n\pi x}{L}\right) - \int\cos\left(\frac{n\pi x}{L}\right)\,dx\right]\right]\Big|_{-L}^{L}$$

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$$A_n = \left[\frac{x^2}{n\pi}\sin\left(\frac{n\pi x}{L}\right) + \frac{2Lx}{(n\pi)^2}\cos\left(\frac{n\pi x}{L}\right) - \frac{2L^2}{(n\pi)^3}\sin\left(\frac{n\pi x}{L}\right)\right]\Bigg|_{-L}^{L}$$

$$A_n = \frac{L^2}{n\pi}\sin(n\pi) + \frac{2L^2}{(n\pi)^2}\cos(n\pi) - \frac{2L^2}{(n\pi)^3}\sin(n\pi)$$

$$-\frac{L^2}{n\pi}\sin(-n\pi) + \frac{2L^2}{(n\pi)^2}\cos(-n\pi) + \frac{2L^2}{(n\pi)^3}\sin(-n\pi)$$

$$A_n = \frac{L^2}{n\pi}\sin(n\pi) + \frac{2L^2}{(n\pi)^2}\cos(n\pi) - \frac{2L^2}{(n\pi)^3}\sin(n\pi)$$

$$+\frac{L^2}{n\pi}\sin(n\pi) + \frac{2L^2}{(n\pi)^2}\cos(n\pi) - \frac{2L^2}{(n\pi)^3}\sin(n\pi)$$

$$A_n = \frac{2L^2}{n\pi}\sin(n\pi) + \frac{4L^2}{(n\pi)^2}\cos(n\pi) - \frac{4L^2}{(n\pi)^3}\sin(n\pi)$$

For $n = 1, 2, 3, \ldots$, $\sin(n\pi) = 0$ and $\cos(n\pi) = (-1)^n$, so the expression for A_n simplifies to

$$A_n = \frac{4L^2}{(n\pi)^2}(-1)^n$$

And for B_n, we get

$$B_n = \frac{1}{L}\int_{-L}^{L} f(x)\sin\left(\frac{n\pi x}{L}\right)\,dx$$

$$B_n = \frac{1}{L}\int_{-L}^{L} x^2\sin\left(\frac{n\pi x}{L}\right)\,dx$$

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The function $f(x) = x^2$ is even, and the sine function is odd, and the product of an even and odd function is an odd function. And since we're integrating over a symmetric interval, the value of the integral has to be 0, so

$$B_n = \frac{1}{L}(0)$$

$$B_n = 0$$

Then the Fourier series representation of $f(x) = x^2$ on $-L \leq x \leq L$ is

$$f(x) = A_0 + \sum_{n=1}^{\infty} A_n \cos\left(\frac{n\pi x}{L}\right) + \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right)$$

$$f(x) = \frac{L^2}{3} + \sum_{n=1}^{\infty} \frac{4L^2}{(n\pi)^2}(-1)^n \cos\left(\frac{n\pi x}{L}\right) + 0$$

$$f(x) = \frac{L^2}{3} + \frac{4L^2}{\pi^2} \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} \cos\left(\frac{n\pi x}{L}\right)$$

Now let's use the Fourier cosine series definition, which means we'll start by calculating A_0 and A_n. For A_0 we get

$$A_0 = \frac{1}{L}\int_0^L f(x)\, dx$$

$$A_0 = \frac{1}{L}\int_0^L x^2\, dx$$

$$A_0 = \frac{1}{3L}x^3\Big|_0^L$$

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$$A_0 = \frac{1}{3L}L^3 - \frac{1}{3L}(0)^3$$

$$A_0 = \frac{L^2}{3}$$

And for A_n we get

$$A_n = \frac{2}{L}\int_0^L f(x)\cos\left(\frac{n\pi x}{L}\right)\,dx$$

$$A_n = \frac{2}{L}\int_0^L x^2\cos\left(\frac{n\pi x}{L}\right)\,dx$$

Use integration by parts with $u = x^2$, $du = 2x\,dx$, $dv = \cos(n\pi x/L)\,dx$, and $v = (L/n\pi)\sin(n\pi x/L)$.

$$A_n = \frac{2}{L}\left[\frac{Lx^2}{n\pi}\sin\left(\frac{n\pi x}{L}\right) - \frac{2L}{n\pi}\int x\sin\left(\frac{n\pi x}{L}\right)\,dx\right]\Bigg|_0^L$$

Use integration by parts with $u = x$, $du = dx$, $dv = \sin(n\pi x/L)\,dx$, and $v = -(L/n\pi)\cos(n\pi x/L)$.

$$A_n = \frac{2}{L}\left[\frac{Lx^2}{n\pi}\sin\left(\frac{n\pi x}{L}\right) - \frac{2L}{n\pi}\left[-\frac{Lx}{n\pi}\cos\left(\frac{n\pi x}{L}\right) + \frac{L}{n\pi}\int\cos\left(\frac{n\pi x}{L}\right)\,dx\right]\right]\Bigg|_0^L$$

$$A_n = \frac{2}{L}\left[\frac{Lx^2}{n\pi}\sin\left(\frac{n\pi x}{L}\right) + \frac{2L^2x}{(n\pi)^2}\cos\left(\frac{n\pi x}{L}\right) - \frac{2L^3}{(n\pi)^3}\sin\left(\frac{n\pi x}{L}\right)\right]\Bigg|_0^L$$

$$A_n = \frac{2x^2}{n\pi}\sin\left(\frac{n\pi x}{L}\right) + \frac{4Lx}{(n\pi)^2}\cos\left(\frac{n\pi x}{L}\right) - \frac{4L^2}{(n\pi)^3}\sin\left(\frac{n\pi x}{L}\right)\Bigg|_0^L$$

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$$A_n = \frac{2L^2}{n\pi}\sin(n\pi) + \frac{4L^2}{(n\pi)^2}\cos(n\pi) - \frac{4L^2}{(n\pi)^3}\sin(n\pi)$$

For $n = 1, 2, 3, \ldots$, $\sin(n\pi) = 0$ and $\cos(n\pi) = (-1)^n$, so the expression for A_n simplifies to

$$A_n = \frac{4L^2}{(n\pi)^2}\cos(n\pi)$$

$$A_n = \frac{4L^2}{(n\pi)^2}(-1)^n$$

Then the Fourier cosine series of $f(x) = x^2$ on $-L \leq x \leq L$ is

$$f(x) = A_0 + \sum_{n=1}^{\infty} A_n \cos\left(\frac{n\pi x}{L}\right)$$

$$f(x) = \frac{L^2}{3} + \sum_{n=1}^{\infty} \frac{4L^2}{(n\pi)^2}(-1)^n \cos\left(\frac{n\pi x}{L}\right)$$

$$f(x) = \frac{L^2}{3} + \frac{4L^2}{\pi^2}\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2}\cos\left(\frac{n\pi x}{L}\right)$$

Because $f(x) = x^2$ is an even function on $-L \leq x \leq L$, we get equivalent representations of $f(x)$ for both its Fourier series and its Fourier cosine series.

Functions that aren't even

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While it's easy to model even functions with a Fourier cosine series, we can also use the cosine series to model functions that aren't even.

We start with the idea that, for any function defined on $0 \leq x \leq L$, we can find its even extension on $-L \leq x \leq 0$. We talked about the even and odd extensions of a function when we looked at periodic extensions earlier. But as a reminder, given $f(x)$ defined on $0 \leq x \leq L$, we create its reflection over the y-axis on $-L \leq x \leq 0$, thereby creating a new function $g(x)$,

$$g(x) = \begin{cases} f(x) & 0 \leq x \leq L \\ f(-x) & -L \leq x < 0 \end{cases}$$

which is itself an even function defined on $-L \leq x \leq L$.

Because $g(x)$ is even, we know we can find the Fourier cosine series representation of $g(x)$ (or its Fourier series representation, in general) on $-L \leq x \leq L$.

But then, because $g(x)$ and $f(x)$ are equivalent on $0 \leq x \leq L$, we know that the Fourier cosine series of $g(x)$ (or the Fourier series of $g(x)$), will be equivalent to the Fourier cosine series of $f(x)$ (or the Fourier series of $f(x)$), as long as we restrict the interval of definition to $0 \leq x \leq L$, instead of $-L \leq x \leq L$.

So let's do another example, this time where we find the Fourier cosine series of a function that isn't even.

Example

Find the Fourier cosine series of $f(x) = x$ on $0 \leq x \leq L$.

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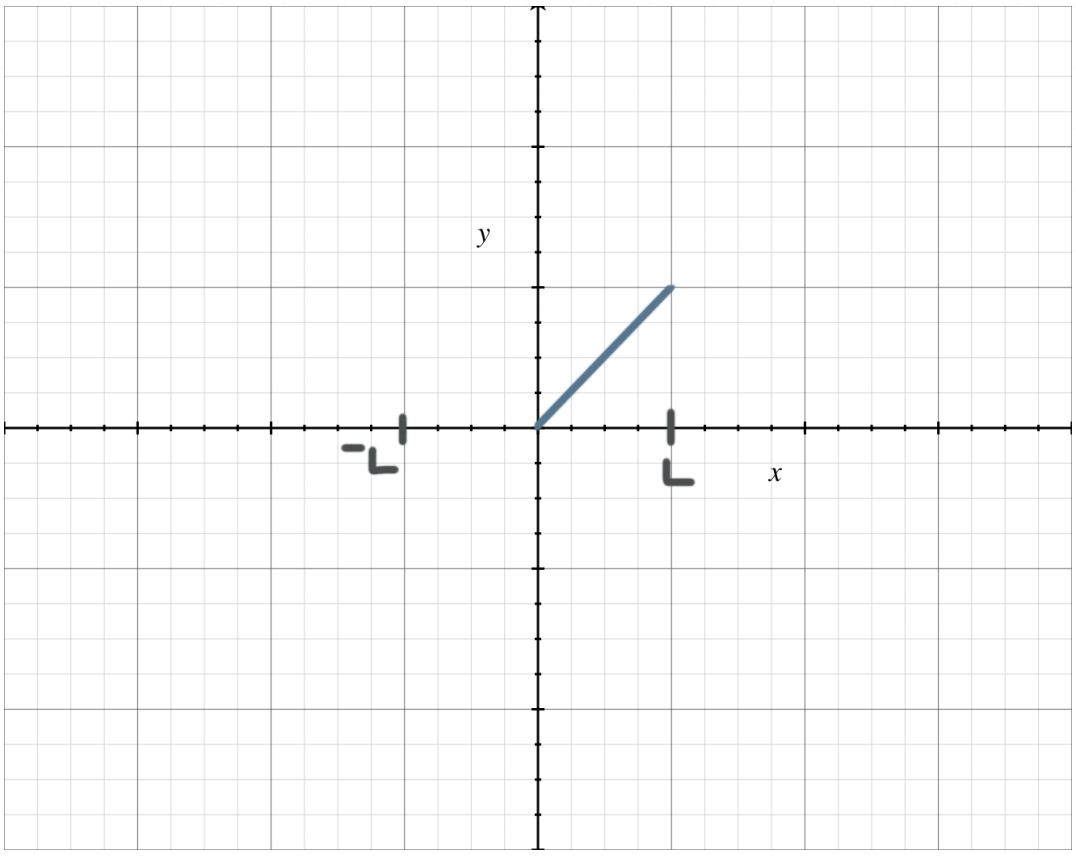


This function is odd, so we'll start by finding the even extension of the odd function.

$$g(x) = \begin{cases} f(x) & 0 \leq x \leq L \\ f(-x) & -L \leq x < 0 \end{cases}$$

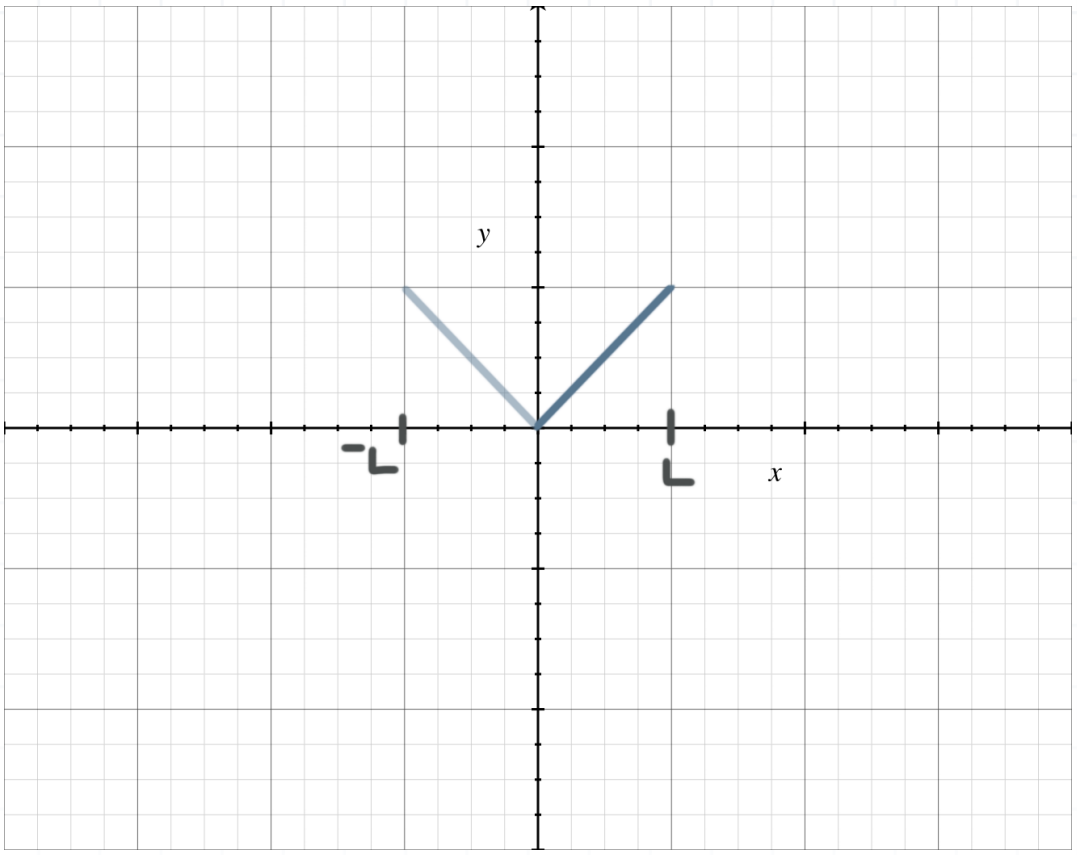
$$g(x) = \begin{cases} x & 0 \leq x \leq L \\ -x & -L \leq x < 0 \end{cases}$$

A sketch of the odd function $f(x)$ on $0 \leq x \leq L$ is



and the sketch of its even extension $g(x)$ on $-L \leq x \leq L$ is

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Because the even extension is an even function, we can find its Fourier cosine series representation, starting with calculating A_0 and A_n. For A_0 we get

$$A_0 = \frac{1}{L}\int_0^L f(x)\ dx$$

$$A_0 = \frac{1}{L}\int_0^L x\ dx$$

$$A_0 = \frac{1}{2L}x^2\Big|_0^L$$

$$A_0 = \frac{1}{2L}(L)^2 - \frac{1}{2L}(0)^2$$

$$A_0 = \frac{L}{2}$$

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And for A_n we get

$$A_n = \frac{2}{L}\int_0^L f(x)\cos\left(\frac{n\pi x}{L}\right)\,dx$$

$$A_n = \frac{2}{L}\int_0^L x\cos\left(\frac{n\pi x}{L}\right)\,dx$$

Use integration by parts with $u = x$, $du = dx$, $dv = \cos(n\pi x/L)\,dx$, and $v = (L/n\pi)\sin(n\pi x/L)$.

$$A_n = \frac{2}{L}\left[\frac{Lx}{n\pi}\sin\left(\frac{n\pi x}{L}\right) - \frac{L}{n\pi}\int \sin\left(\frac{n\pi x}{L}\right)\,dx\right]\Bigg|_0^L$$

$$A_n = \frac{2}{L}\left[\frac{Lx}{n\pi}\sin\left(\frac{n\pi x}{L}\right) + \frac{L^2}{(n\pi)^2}\cos\left(\frac{n\pi x}{L}\right)\right]\Bigg|_0^L$$

$$A_n = \frac{2x}{n\pi}\sin\left(\frac{n\pi x}{L}\right) + \frac{2L}{(n\pi)^2}\cos\left(\frac{n\pi x}{L}\right)\Bigg|_0^L$$

$$A_n = \frac{2L}{n\pi}\sin(n\pi) + \frac{2L}{(n\pi)^2}\cos(n\pi) - \frac{2L}{(n\pi)^2}$$

For $n = 1, 2, 3, \ldots$, $\sin(n\pi) = 0$ and $\cos(n\pi) = (-1)^n$, so the expression for A_n simplifies to

$$A_n = \frac{2L}{n\pi}(0) + \frac{2L}{(n\pi)^2}(-1)^n - \frac{2L}{(n\pi)^2}$$

$$A_n = \frac{2L((-1)^n - 1)}{(n\pi)^2}$$

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Then the Fourier cosine series of $f(x) = x$ on $0 \leq x \leq L$ is

$$f(x) = A_0 + \sum_{n=1}^{\infty} A_n \cos\left(\frac{n\pi x}{L}\right)$$

$$f(x) = \frac{L}{2} + \sum_{n=1}^{\infty} \frac{2L((-1)^n - 1)}{(n\pi)^2} \cos\left(\frac{n\pi x}{L}\right)$$

$$f(x) = \frac{L}{2} + \frac{2L}{\pi^2} \sum_{n=1}^{\infty} \frac{(-1)^n - 1}{n^2} \cos\left(\frac{n\pi x}{L}\right)$$

Even though the function $f(x) = x$ isn't even, we were still able to find a Fourier cosine series to represent it on $0 \leq x \leq L$.

Convergence of the cosine series

We know from earlier that a Fourier series will converge to the function $f(x)$ that it represents as long as $f(x)$ is piecewise smooth and its periodic extension is continuous.

Otherwise the Fourier series will converge to the average of the one-sided limits if the periodic extension has a jump discontinuity between periods.

But now we want to specifically address the convergence of a Fourier cosine series.

If we're looking at the Fourier cosine series of an even function on $-L \leq x \leq L$, then it'll converge to $f(x)$ just like a normal Fourier series. Of

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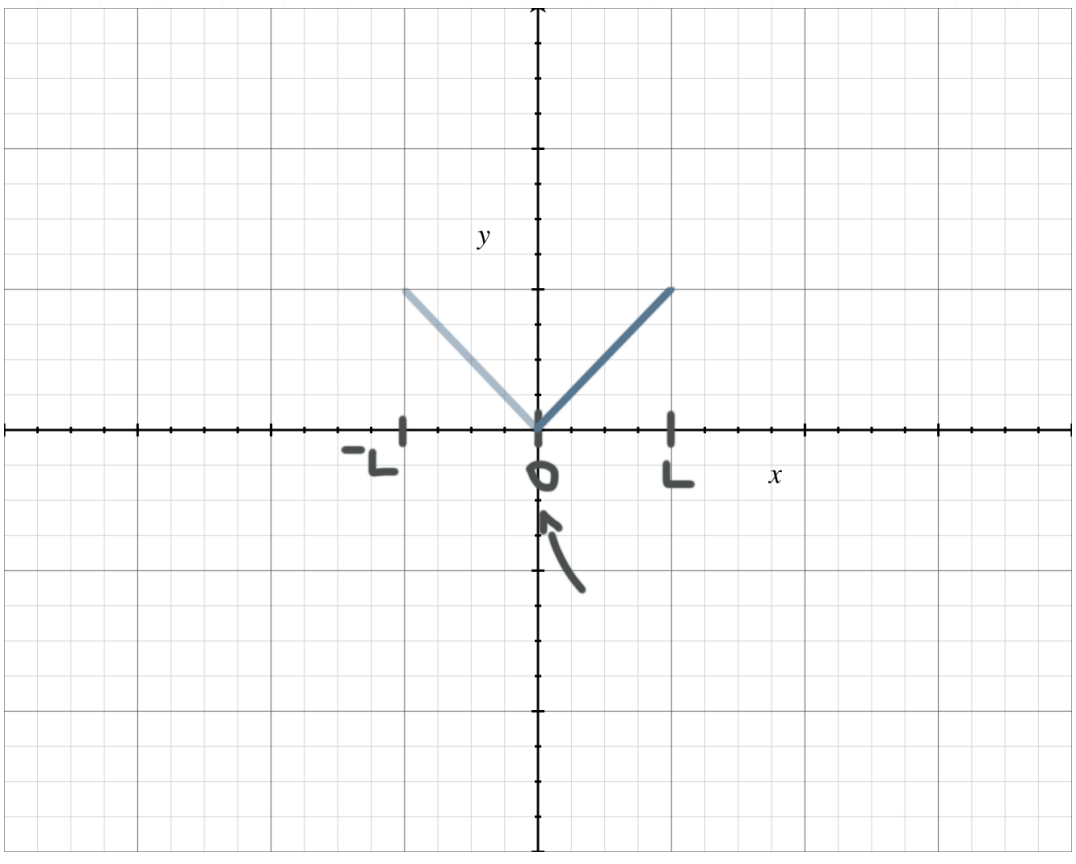


course, that's because the Fourier cosine series of an even function is just the special case of the general Fourier series, where $B_n = 0$ and the sine series part of the Fourier series disappears.

But what about the convergence of the Fourier cosine series of a function that isn't even?

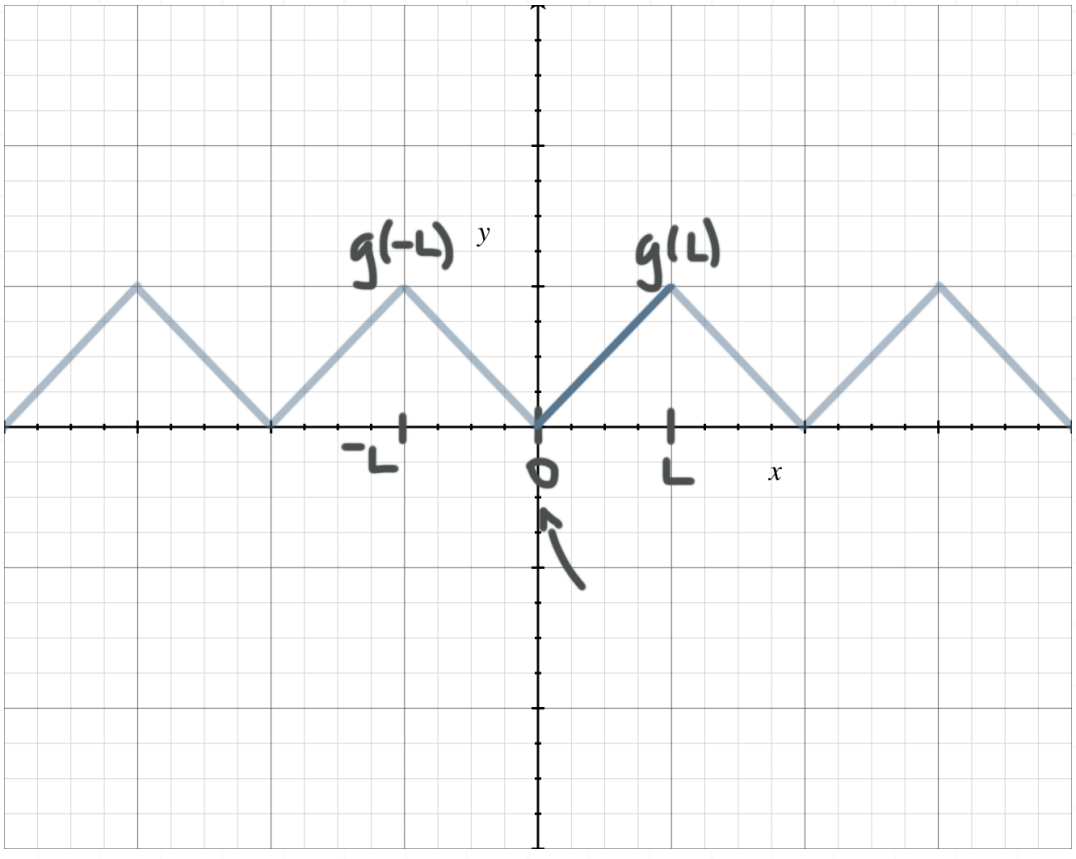
Well remember, if we take a function defined on $0 \leq x \leq L$ and build its even extension, that even extension is now an even function $g(x)$ on $-L \leq x \leq L$, and we can therefore find the Fourier cosine series of that even $g(x)$ function on $-L \leq x \leq L$.

The original function and its even extension meet each other at $x = 0$,



and if we build the periodic extension of that even extension,

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then we see that the periodic extension is continuous between periods, since $g(-L) = g(L)$.

Which means, based on what we already know about the convergence of a Fourier series, that the Fourier series representation of $g(x)$ will converge to $g(x)$ on $-L \leq x \leq L$. The Fourier cosine series, as a special case of the Fourier series, will also converge to $g(x)$ on $-L \leq x \leq L$.

And because the even extension $g(x)$ is equivalent to the original function $f(x)$ on $0 \leq x \leq L$, we can conclude that the Fourier cosine series representation of $f(x)$ will converge to $f(x)$ on $0 \leq x \leq L$.

Therefore,

- the Fourier cosine series of an even function $f(x)$ will converge to $f(x)$ on $-L \leq x \leq L$

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- the Fourier cosine series of a function $f(x)$ that isn't even will converge to $f(x)$ on $0 \leq x \leq L$

The only exception is if the function $f(x)$ contains a jump discontinuity somewhere within $0 \leq x \leq L$. But just like with the general Fourier series, the Fourier cosine series will converge to the average of the one-sided limits at that jump discontinuity.

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